Digital Image Processing – Course Project

Real-Time Facial Analysis System
Role: Team Leader & Error Handling Lead
Inha University | Sept 2025 – Dec 2025

Project Overview

The project focused on designing and implementing a **real-time facial analysis pipeline** capable of detecting faces, extracting facial features, and classifying emotions from live video streams. The system was developed using **OpenCV** for image acquisition and preprocessing, and **DeepFace** for deep learning–based facial feature extraction and emotion recognition.

Role and Key Contributions

As the **team leader**, I was responsible for both system architecture and technical integration across modules.

- ✓ Designed the **end-to-end pipeline architecture**, defining clear interfaces between face detection, preprocessing, feature extraction, and emotion classification stages.
- ✓ Implemented the **live video processing module** using OpenCV, handling frame capture, preprocessing, and visualization in real time.
- ✓ Integrated **DeepFace pretrained models** for facial feature extraction and emotion classification, adapting them for continuous video input rather than static images.
- ✓ Led system integration and testing, resolving conflicts between independently developed components.
- ✓ Coordinated team workflows, reviewed code for correctness and robustness, and ensured the final system met performance requirements.

Technical Challenges and Solutions

1. Inconsistent Face Detection and Alignment

Challenge:
Face detection accuracy degraded under poor lighting conditions and varied head poses.

Solution:

- Applied OpenCV-based preprocessing techniques such as grayscale conversion, histogram equalization, and normalization.
- Added face alignment and bounding-box validation to ensure consistent ROI extraction prior to DeepFace inference.

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2. Multiple Face Handling and ROI Errors

Challenge:
Incorrect emotion classification occurred when multiple faces appeared simultaneously in a frame.

Solution:

- Implemented confidence thresholding and non-maximum suppression logic to filter unreliable detections.
- Ensured each detected face was processed independently with correctly mapped emotion labels.

Outcome

The final system successfully performed **real-time face detection and emotion analysis** on live webcam video, demonstrating a practical application of digital image processing concepts and deep learning integration. The project highlighted challenges in real-time computer vision systems, particularly in performance optimization, robustness, and system-level debugging.